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ITAI 4370: AI 5/6G Comm & ORAN Net
Assignment 03
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Question 1 - *AI/ML Applications in RAN: Give one example of how AI/ML can improve radio resource management and one example of how it can enhance network reliability in Open RAN.*

AI and machine learning let the radio access network make faster and smarter decisions than fixed rules can. For radio resource management, a model learns from past and live traffic and moves limited resources such as spectrum and bandwidth to where they are needed before congestion builds up. For reliability, that same learning watches the network and its equipment for early warning signs, so problems are caught and worked around before they grow into an outage (IEEE 5G/6G Innovation Testbed, n.d.). In Open RAN, decisions like these usually run as an app on the RAN Intelligent Controller, the part of the network that adds this intelligent, software-driven control (VIAVI Solutions, 2022).

Examples:

- For radio resource management, a model forecasts a busy period and moves extra spectrum to a cell before the traffic rises.
- For network reliability, a model flags a base station that is starting to fail so it can be fixed before service drops.

Question 2 - *SON and Predictive Maintenance: Describe how self-optimization and self-healing work in Self-Organizing Networks (SON). How does predictive maintenance improve Quality of Service (QoS)?*

A Self-Organizing Network, or SON, runs many of its own management jobs automatically so operators do not have to handle them by hand. Self-optimization is the network constantly tuning its own settings, such as power levels, handover thresholds, and traffic balancing, as conditions change, which keeps service quality steady. Self-healing is the network spotting faults on its own and working around them, for example by having nearby cells adjust their coverage when one cell fails (TechTarget, n.d.). Predictive maintenance builds on this by using data to predict failures before they happen, so faults are fixed early instead of after they cause harm, which protects Quality of Service by avoiding sudden outages and slowdowns (Mobara et al., 2025).

Examples:

- Self-optimization can adjust handover settings and balance traffic across cells during a rush hour so calls do not drop.
- Self-healing can have neighboring cells widen their coverage when one cell goes down, so users keep service until it is repaired.

Question 3 - *Hands-On Lab Application: In the Python-based resource allocation simulation lab, what does normalizing user demand achieve? How does this relate to real-world resource scheduling in Open RAN?*

In the lab, the code first weights each slice's demand by priority and then normalizes it, dividing each slice's weighted demand by the total and scaling the result to the fixed pool of 500 units. Normalizing turns raw demands of different sizes into fair, proportional shares that always add up to the capacity that is actually available, so no single slice can take more than its weighted share. This mirrors how a real network shares a limited pool of radio resources among its slices and users, giving priority, latency-sensitive traffic the larger share while staying within the total capacity it has (NETSCOUT, n.d.).

Examples:

- Three slices asking for different amounts are scaled so their allocations add up to exactly the 500 units available.
- A real network gives latency-sensitive traffic the larger share of limited spectrum, the same way the lab weights favor URLLC.

Question 4 - *Traditional RAN vs. Open RAN: Explain two key differences between traditional RAN and Open RAN architectures. How does Open RAN promote vendor interoperability?*

A traditional RAN comes from a single vendor as one tightly bundled package of hardware and software, with closed and proprietary links between its parts, which tends to lock the operator into that one supplier. Open RAN breaks that bundle into separate pieces that connect over open and standard interfaces, and it runs much of the RAN as software on common hardware. Because those interfaces are open and standard, an operator can mix equipment from different vendors instead of being tied to one, and that is how Open RAN promotes vendor interoperability (RCR Wireless News, 2026).

Examples:

- A traditional RAN uses closed, proprietary links between its parts, while Open RAN uses open, standard interfaces.
- A traditional RAN runs on purpose-built vendor hardware, while Open RAN runs much of its work as software on common hardware.

Question 5 - *RAN Intelligent Controller (RIC): Distinguish between the roles of the Near-RT RIC and Non-RT RIC. Provide an example function for each.*

The RAN Intelligent Controller, or RIC, is the part of Open RAN that adds smart, software-driven control, and it comes in two forms that are split by speed. The Non-RT RIC works on the slower side, about a second or more, and sits in the management layer where it handles the big-picture work such as setting policies and training the AI models that guide the network. The Near-RT RIC works fast, from roughly ten milliseconds to a second, and sits closer to the radios where it makes quick control decisions on the live network (VIAVI Solutions, 2022).

Examples:

- The Non-RT RIC can train a machine learning model and set a policy that defines quality targets for a group of users.
- The Near-RT RIC can run an app that manages handovers or controls interference in near real time.

References

IEEE 5G/6G Innovation Testbed. (n.d.). Why is machine learning important for 5G wireless? https://testbed.ieee.org/why-is-machine-learning-important-for-5g-wireless/

Mobara, P., Moskolaï Ngossaha, J., Djeumen Djatcha, R., Tchappi, I., & Bowong Tsakou, S. (2025). AI-driven load balancing in paravirtualized 5G networks: Architecture, implementation, and future prospects. Procedia Computer Science, 257, 599-606. https://doi.org/10.1016/j.procs.2025.03.077

NETSCOUT. (n.d.). What is network slicing? https://www.netscout.com/blog/what-network-slicing

RCR Wireless News. (2026, April 8). Open interfaces, smarter networks: The O-RAN advantagefor Indoor Deployments. https://www.rcrwireless.com/20260408/open_ran/open-interfaces-o-ran

TechTarget. (n.d.). What is a self-organizing network (SON) and do you need one? https://www.techtarget.com/searchnetworking/definition/What-is-a-self-organizing-network

VIAVI Solutions. (2022, May 6). RAN Intelligent Controller: Near-RT and non-RT. https://blog.viavisolutions.com/2022/05/06/ran-intelligent-real-time-and-non-real-time-ric-the-test-challenges/